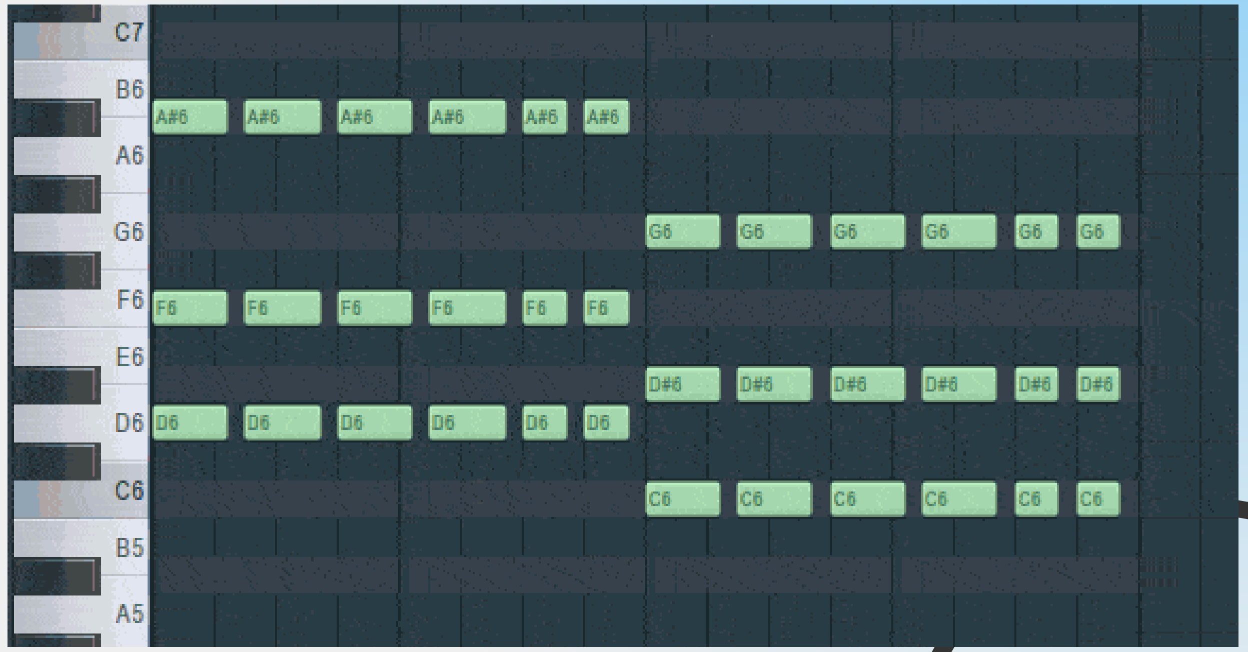
# Audio2PianoRoll




Francesco Brigante, 1987197

# Motivations

## Education and Analysis

PianoRoll is a representation rich of information and it provides an easy way to read music
The academic literature on guitar transcription is limited compared to the research available on piano transcription

## Automatic Music Transcription

PianoRoll can be easily converted to MIDI format for playing and it can be used as an intermediate step for AMT (Both for sheet music and guitar tabs)

## Feature Extraction

Baseline model can be used as a musical feature extractor for building new applications

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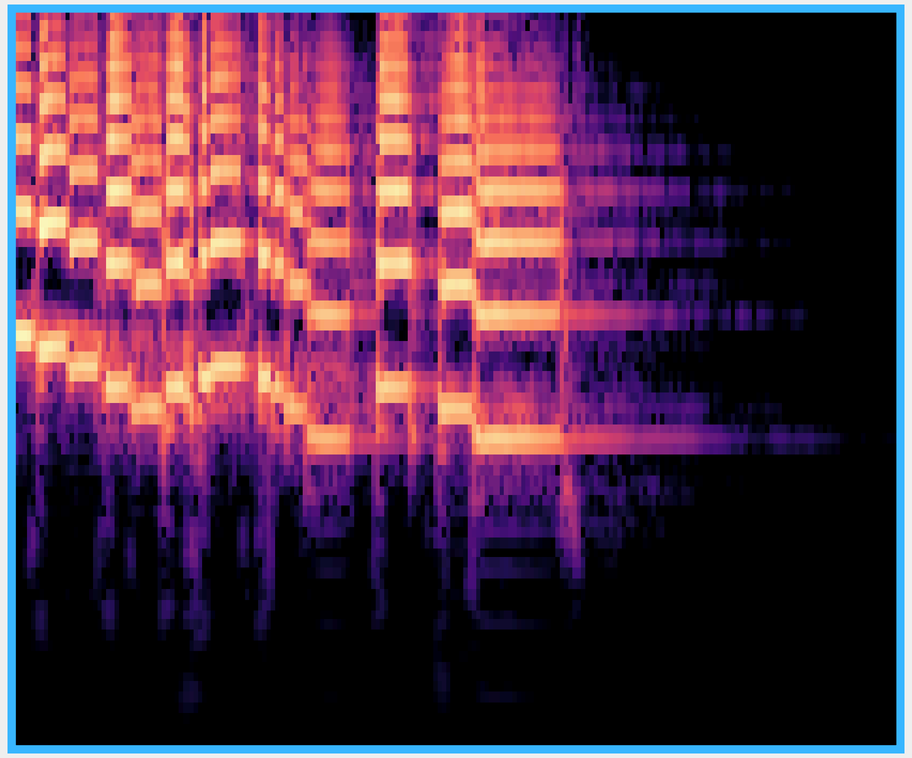
# Dataset & Input Representation

The dataset used was **GuitarSet**:

- 360 audios * 4 recording modalities -> **Data Augmentation**
- Various musical genres
- Rich annotations (pitch, MIDI, tempo, guitar fret, ...)



**Constant-Q Transform:**

- Rich representation
- Better mapping between frequency bins and MIDI notes
- Musical scales are based on logarithmic intervals and the CQT spaces frequency bins logarithmically

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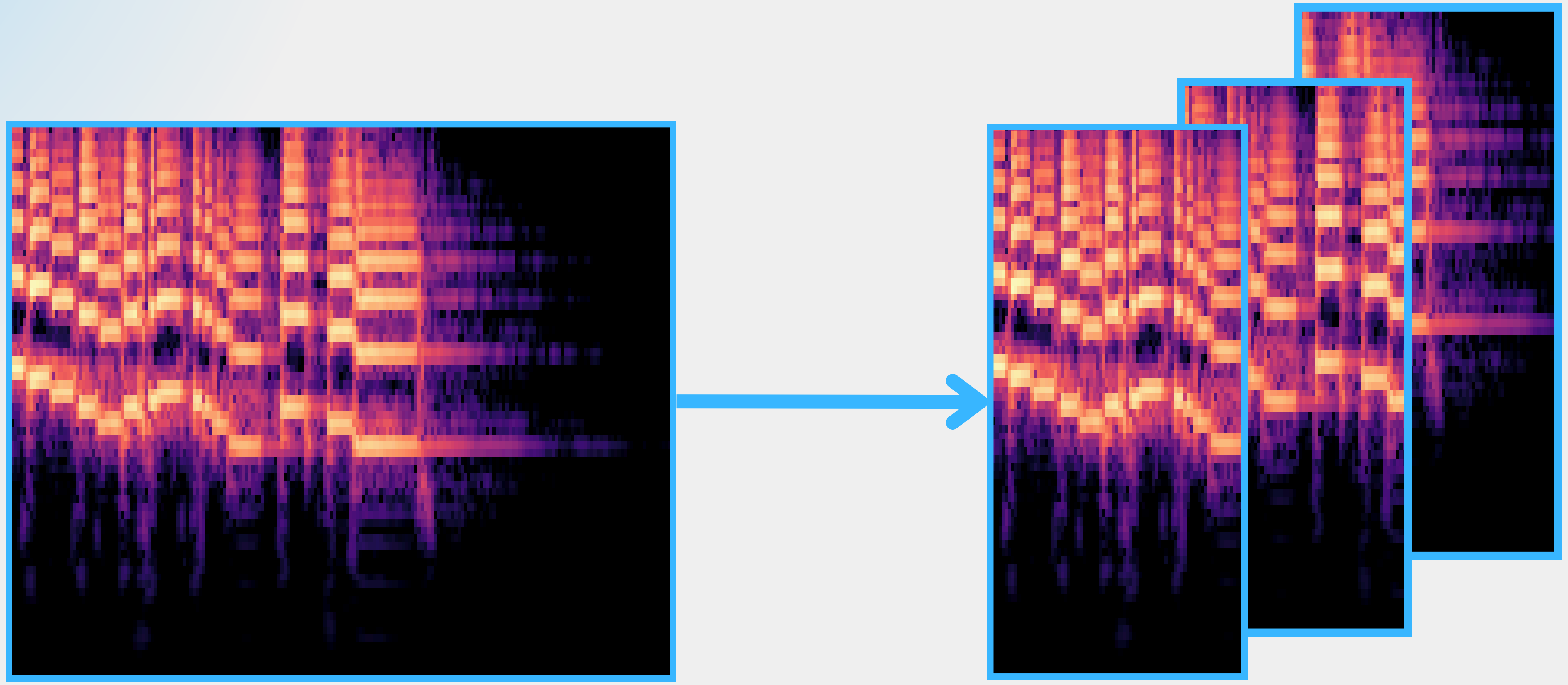
# How to deal with different lengths?



CQT is divided into sections with an overlapping factor of 50%

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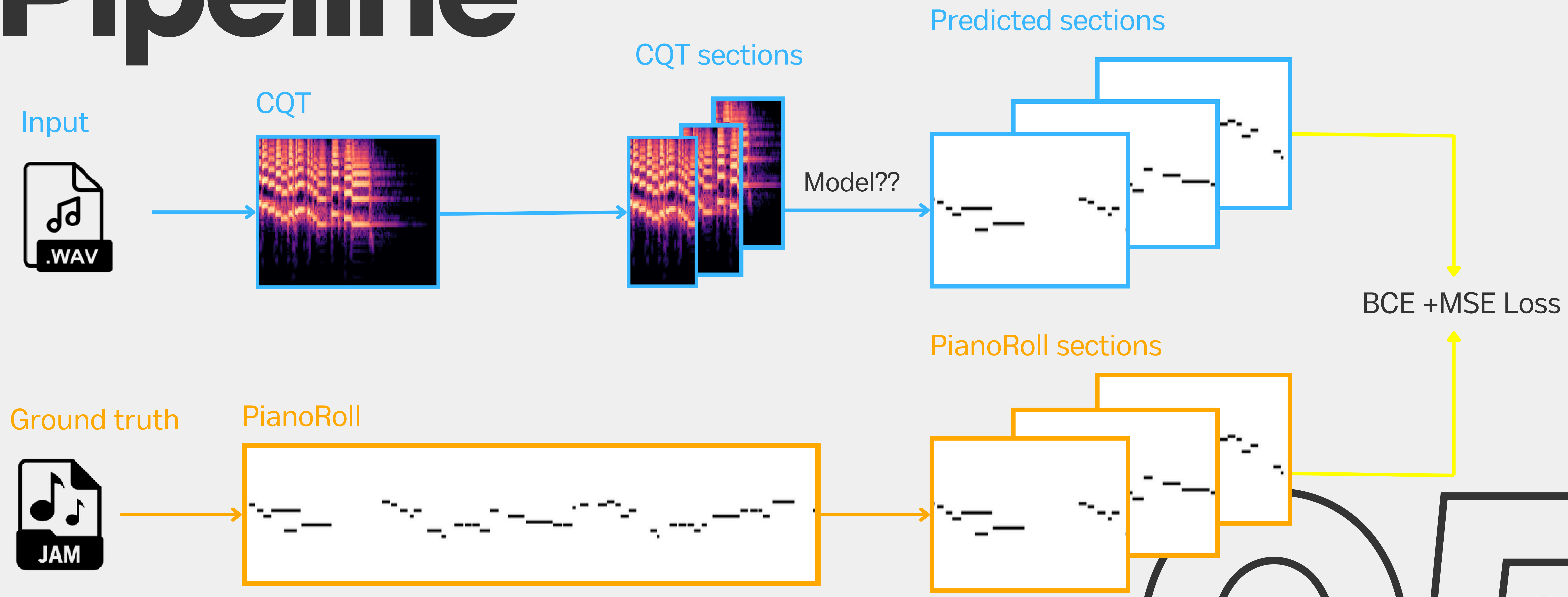
# Pipeline


Predicted sections
CQT sections
Input
CQT
.WAV
Model??
BCE +MSE Loss
PianoRoll sections
Ground truth
PianoRoll
JAM

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# Model: U-Net

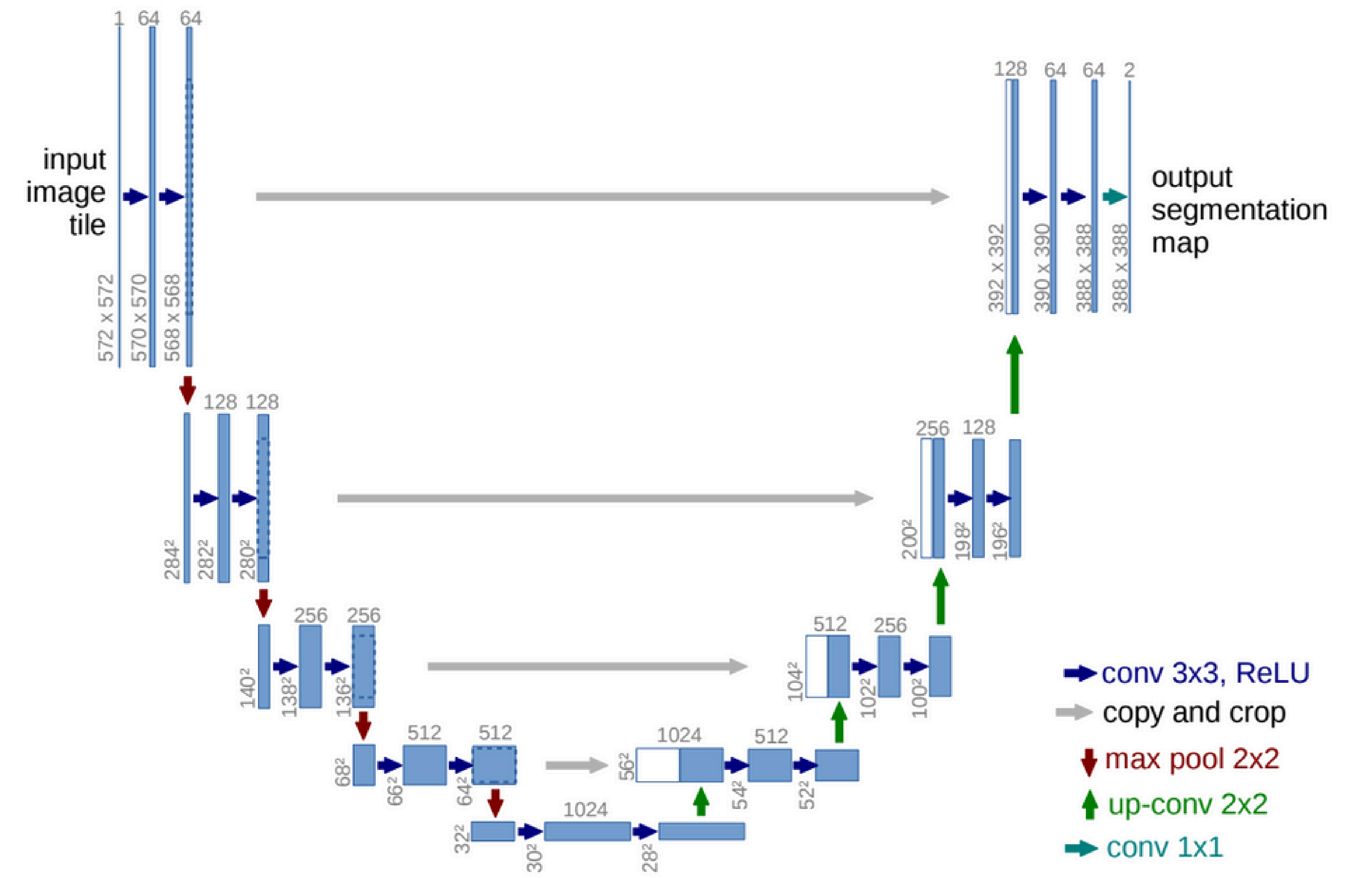


**Fig. 1.** U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

- Originally used for Biomedical Image Segmentation
- Based on a **U-Shape** architecture:
  - Encoder
  - Decoder
  - Skip connections
- Reaches >90% Accuracy in many Segmentation tasks

U-Net: Convolutional Networks for Biomedical Image Segmentation

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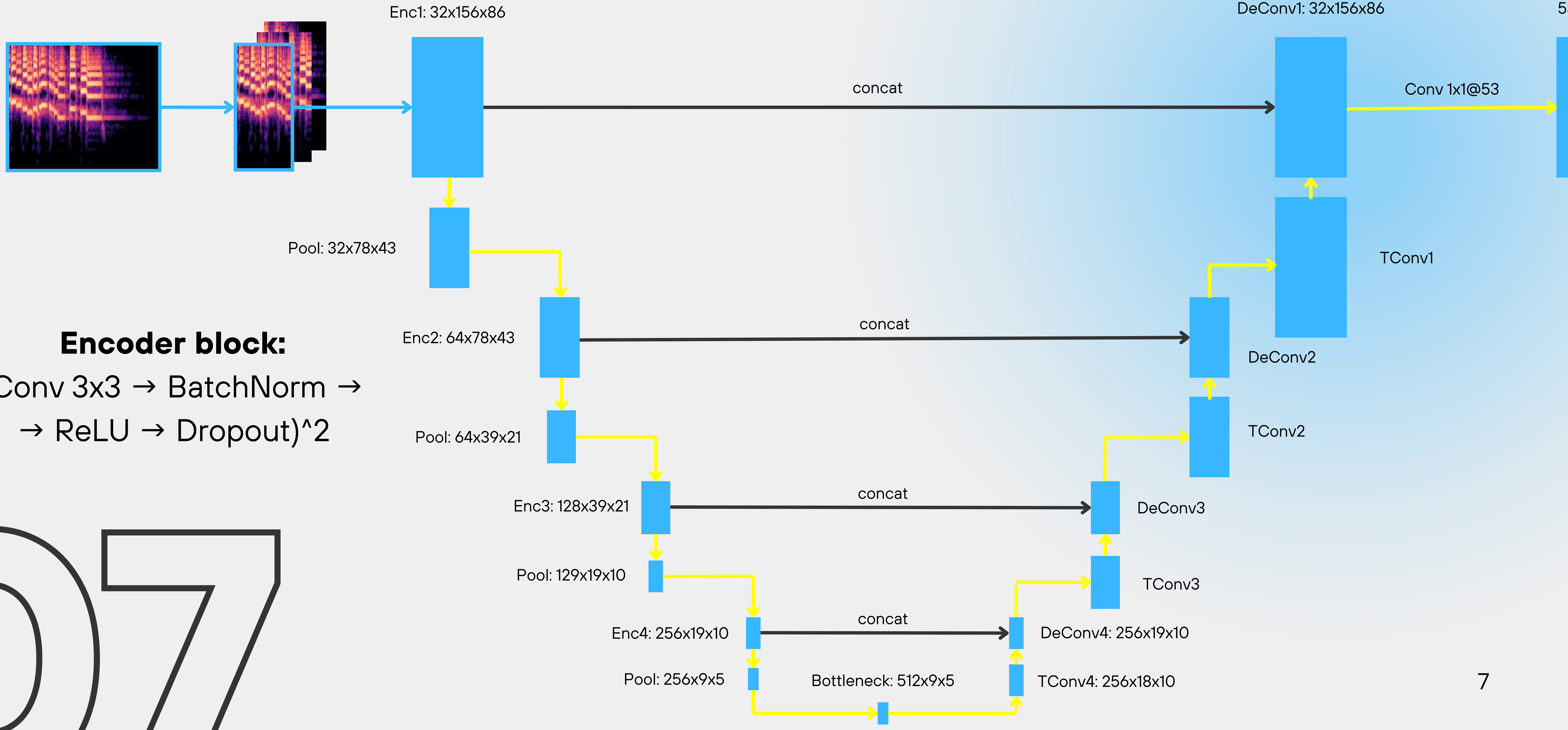


**Encoder block:**

(Conv 3x3 → BatchNorm →

→ ReLU → Dropout)^2

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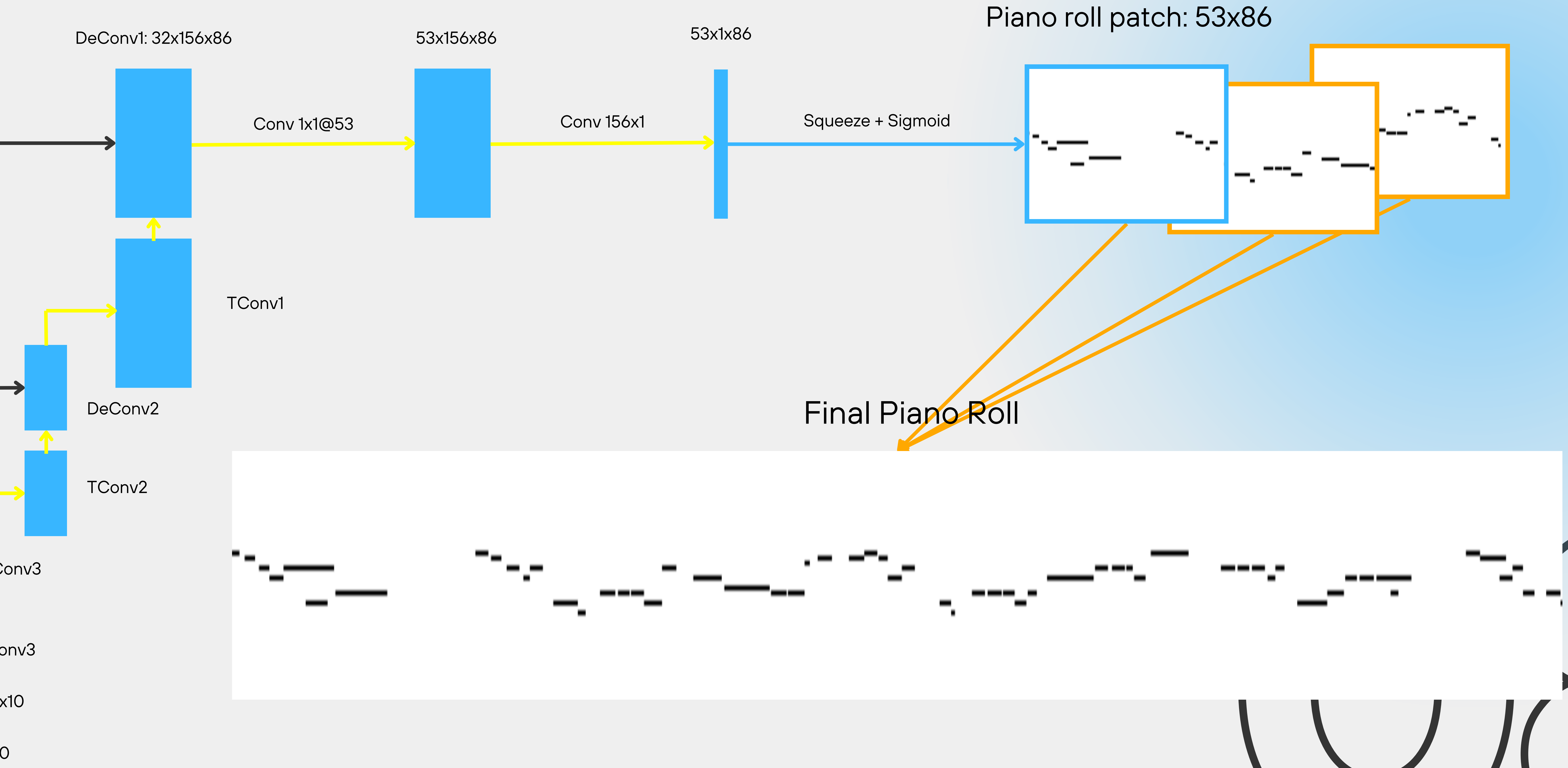
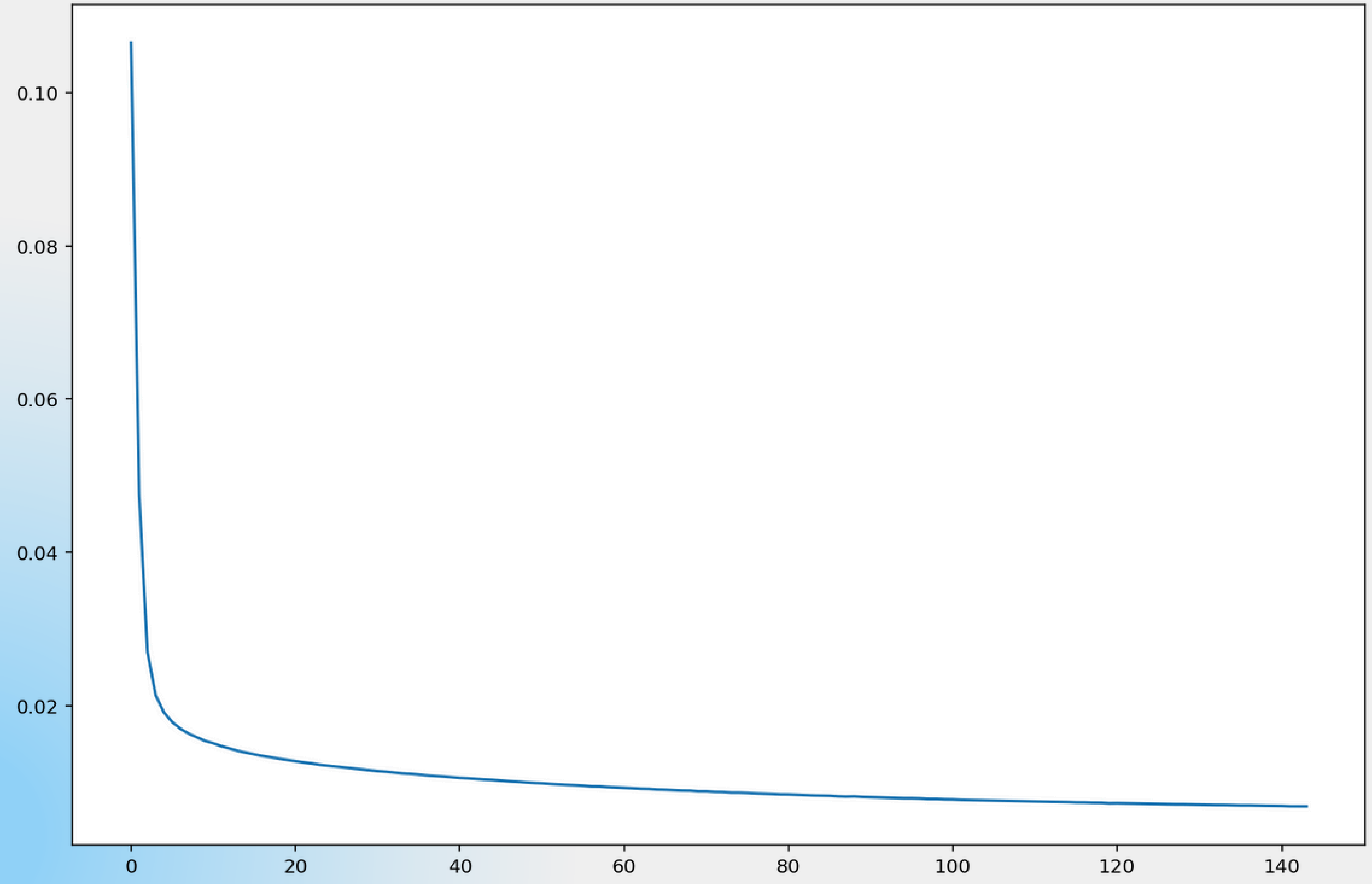

DeConv1: 32x156x86
53x156x86
53x1x86
Piano roll patch: 53x86
Conv 1x1@53
Conv 156x1
Squeeze + Sigmoid
TConv1
DeConv2
TConv2
Final Piano Roll
Conv3
onv3
x10
0

# Training

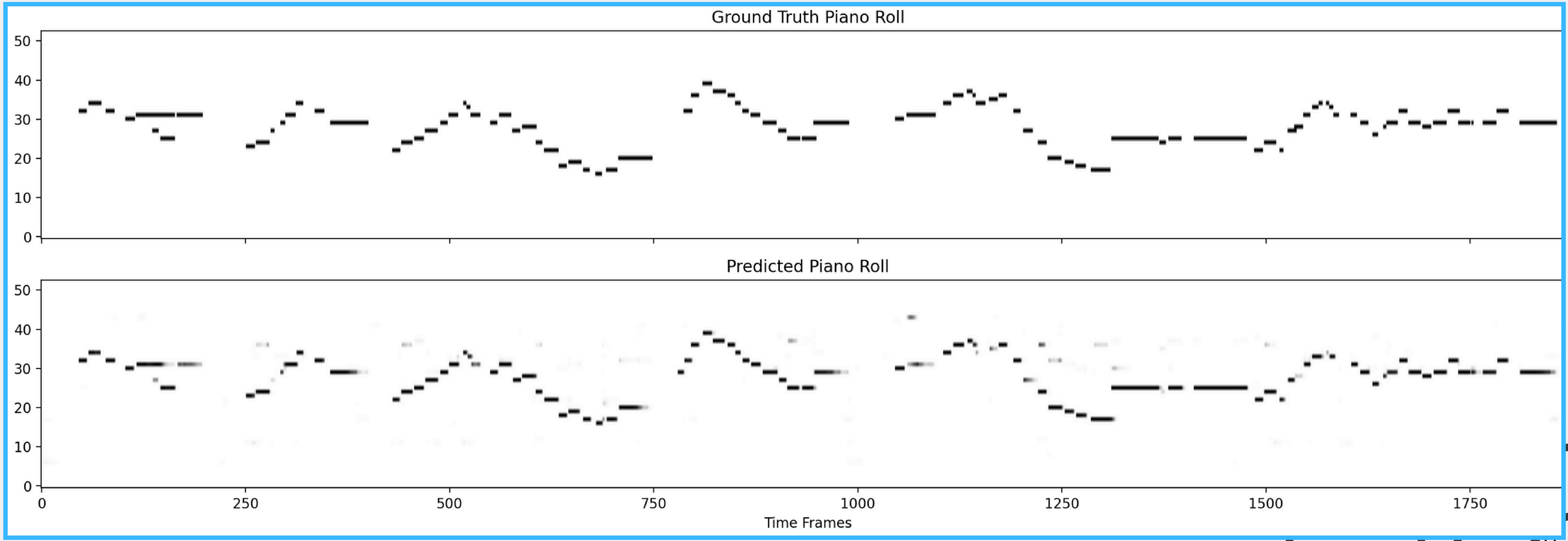





- **Weight initialization:**
  - He initialization for Conv layers (ReLU)
  - Xavier initialization for linear layers and transposed convolutions
- Dataset divided into train(80%) and test(20%)
- Trained for **144 epochs** having **final loss 0.0069**

# Results

- **F1:** 83%
- **Precision:** 85%
- **Recall:** 82%



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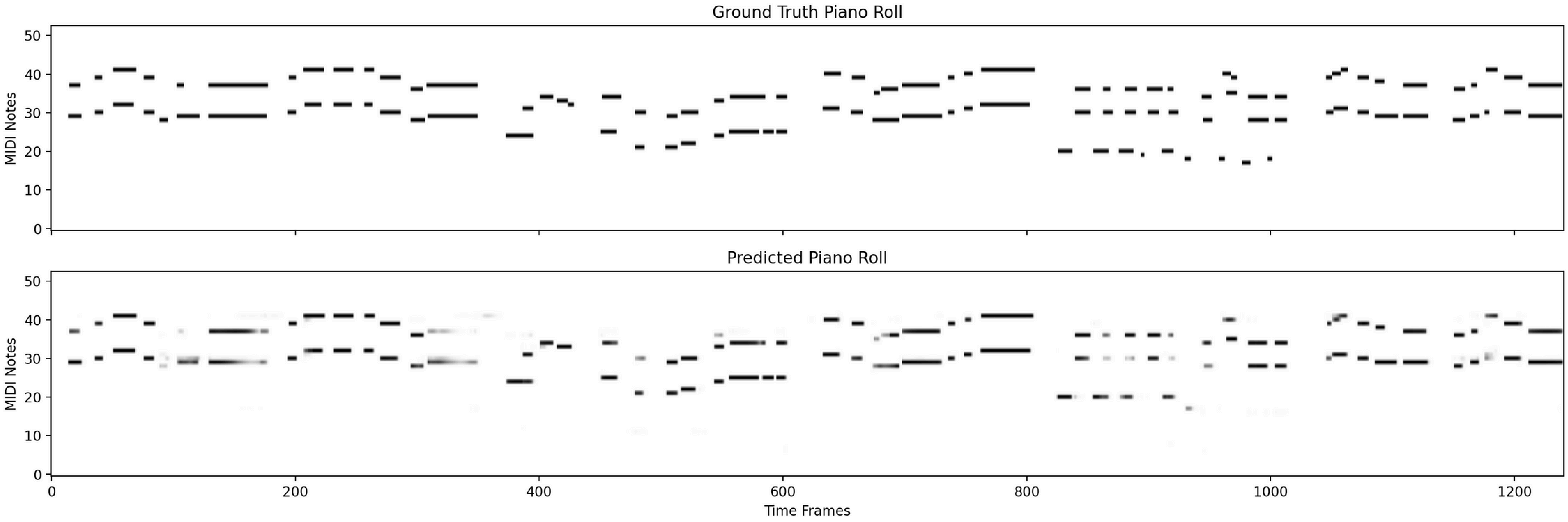

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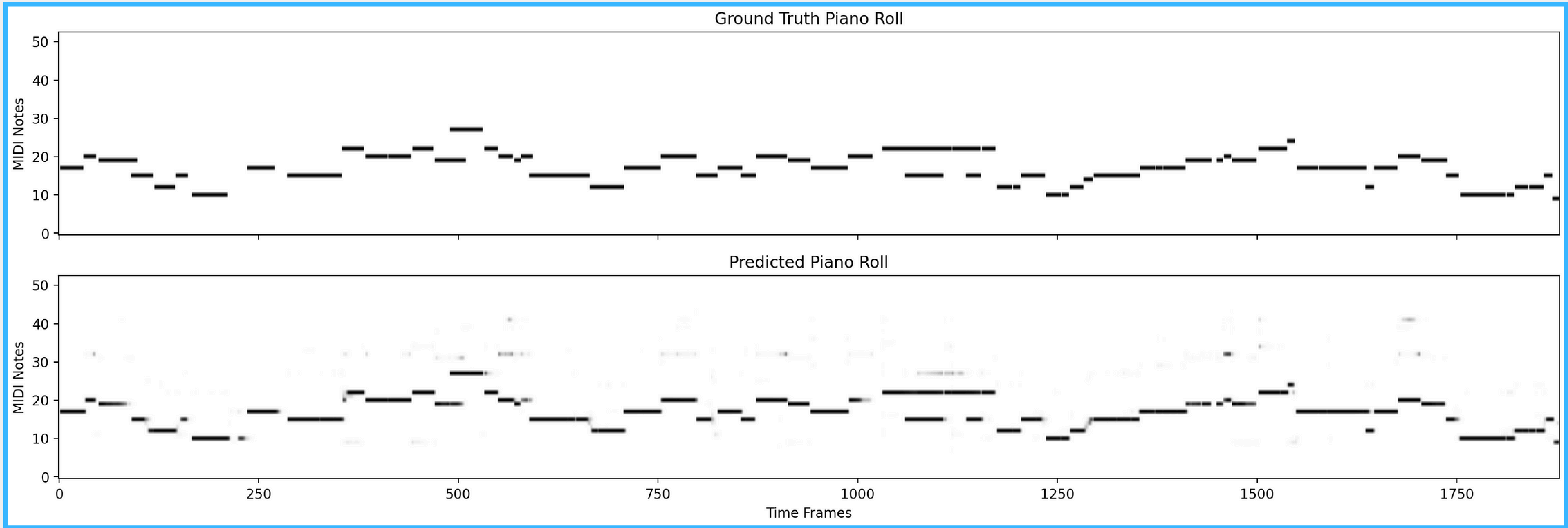

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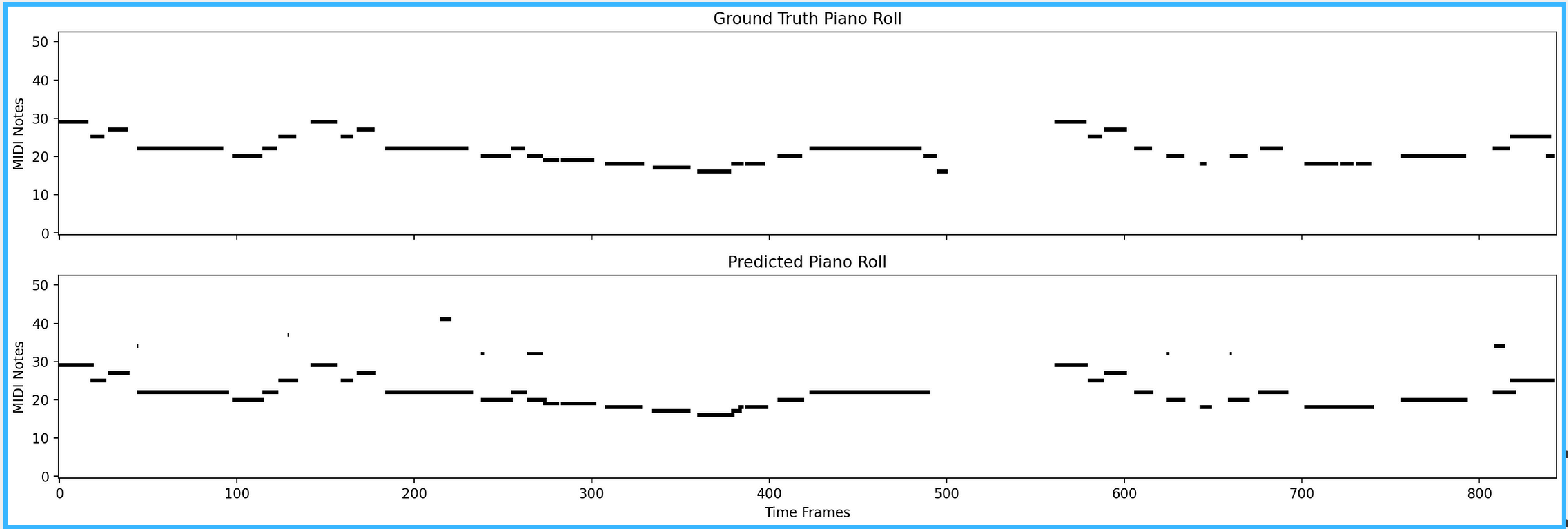

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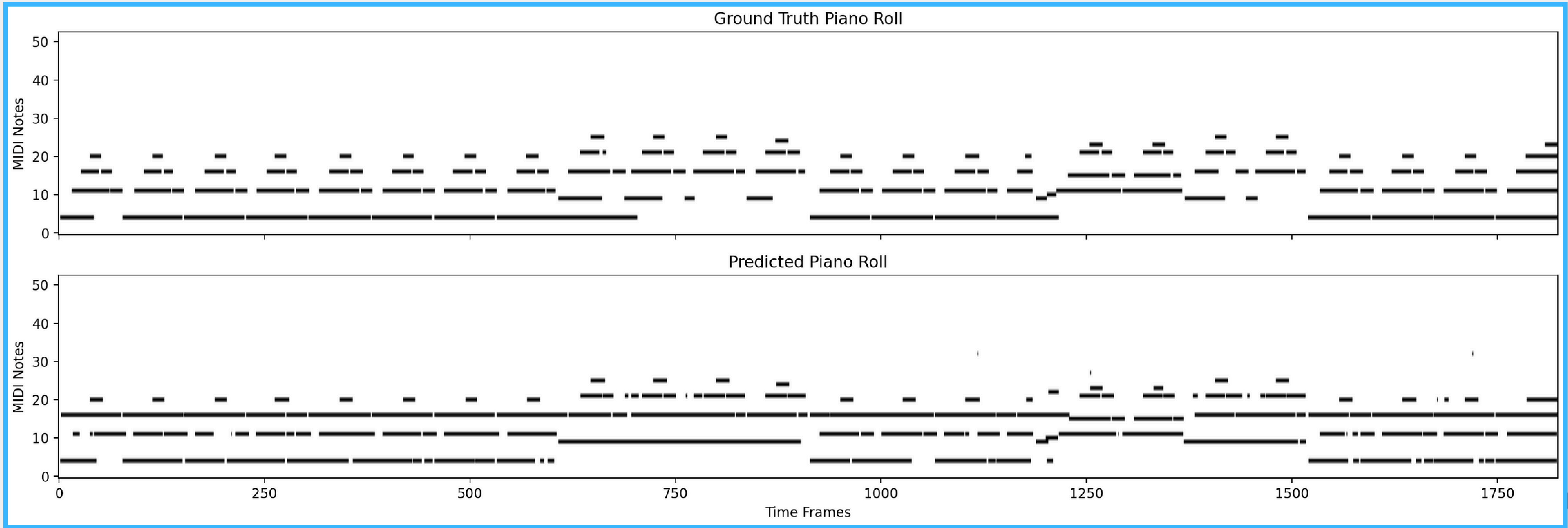

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# Future Work

## Transcription

We can leverage the piano roll representation to transcribe notes both in sheet music and guitar tabs

## Cleaning Predictions

Add post-processing techniques to clean the thresholded predicted piano rolls

## Adding new instruments

The application could be adapted to handle different instruments or more than one instrument per song

## More testing

It could be interesting to keep testing the model with different parameters (e.g. size of overlapping window)

# Thank You!